

Course syllabus for

Academic year 2025/2026 >

TIF345 - Advanced simulation and machine learning

Avancerad simulering och maskininlärning

Course syllabus adopted 2025-02-20 by Head of Programme (or corresponding)

Owner: [MPPHS](#)

7,5 Credits

Grading: TH - Pass with distinction (5), Pass with credit (4), Pass (3), Fail

Education cycle: Second-cycle

Main field of study: Engineering Physics

Department: 16 - PHYSICS

Course round 1

Teaching language: English

Application code: 85114

Open for exchange students: Yes

Block schedule: [A](#)

Maximum participants: 60 (at least 10% of the places are reserved for exchange students)

Credit distribution

Module	Sp1	Sp2	Sp3	Sp4	Summer course [!]	No Sp	Examination dates [!]
0120 Project 7,5 c Grading: TH		7,5 c					

In programs

[MPPHS Physics, Year 1 \(compulsory elective\)](#)

[MPPHS Physics, Year 2 \(elective\)](#)

Examiner:

[Paul Erhart](#)

 [Go to course homepage](#)

Eligibility

General entry requirements for Master's level (second cycle)

Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Specific entry requirements

English 6 (or by other approved means with the equivalent proficiency level)

Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Course specific prerequisites

TIF285 - Learning from data and

FKA121 - Computational physics or equivalent

Aim

The aim of this course is to provide students with in-depth knowledge about statistical methods and machine learning algorithms, and how they can be used to analyze and simulate physical systems in a scientific manner. The course is based on a set of projects, which are accompanied by lectures, and hands-on computer exercises. During the course, the students will be exposed to advanced scientific research problems, with the aim of reproducing state-of-the-art scientific results. The students will use e.g. the Python programming language and relevant open-source libraries, and will learn to develop and structure computer codes for carrying out scientific and statistical data analyses.

Learning outcomes (after completion of the course the student should be able to)

- critically examine the description of systems in the physical sciences by different mathematical models
- rationalize the numerical representation of such models at multiple levels of sophistication
- employ statistical inference and machine learning (ML) methods to evaluate and compare models

- explain, using appropriate terminology, methods from ML and statistical inference
- analyze data and write code in scientific and ethical fashion

Content

Advanced simulations in the physical sciences can benefit from ML methods in multiple ways:

- Uncertainty quantification via Bayesian inference
- Representation of mathematical models via ML models, e.g., neural networks and Gaussian processes
- Parametrization and selection of ML models via regression techniques

with the following subtopics

- Dimensionality reduction and descriptors for physical systems
- Bayesian inference and model selection
- Generalized linear models including Gaussian processes
- Advanced regression and regularization techniques
- Neural networks

All of these aspects will be introduced and examined in the context of modelling in the physical sciences.

Organisation

- Lectures
- Supervised computational exercises (group work on computational projects)
- Selected number of small hand-in assignments
- Computational projects with written reports

Examination including compulsory elements

The final grade is based on the combined performance on hand-in assignments and computational projects.

The course examiner may assess individual students in other ways than what is stated above if there are special reasons for doing so, for example if a student has a decision from Chalmers about disability study support.